Practice Assignment

Transactions · Joins · Indexing · Query Optimization

Part – 1

```
CREATE TABLE customers (     id INT
PRIMARY KEY AUTO_INCREMENT,     name
VARCHAR(100),     email VARCHAR(100)
UNIQUE,     signup_date DATE
);

CREATE TABLE restaurants (     id INT
PRIMARY KEY AUTO_INCREMENT,     name
VARCHAR(100),     cuisine VARCHAR(50)
);

CREATE TABLE orders (     id INT PRIMARY
KEY AUTO_INCREMENT,     customer_id
INT,     restaurant_id INT,     total
DECIMAL(10,2),     status VARCHAR(20),
created_at DATETIME,

   FOREIGN KEY (customer_id) REFERENCES customers(id),
   FOREIGN KEY (restaurant_id) REFERENCES restaurants(id)
);

CREATE TABLE accounts (     id INT
PRIMARY KEY AUTO_INCREMENT,     name
VARCHAR(100),
   balance DECIMAL(10,2)
);


INSERT INTO customers (name, email, signup_date) VALUES
```

('saad, “saad@gmail.com', '2026-01-05'),

('Aditi Sharma', 'aditi.sharma@gmail.com', '2026-01-08'),

('Rohan Patel', 'rohan.patel@gmail.com', '2026-01-12'),

('Priyanshi Joshi', 'priyanshi.joshi@gmail.com', '2026-01-18'),

('Dhruv Shah', 'dhruv.shah@gmail.com', '2026-01-25'),

('Khushi Mehta', 'khushi.mehta@gmail.com', '2026-02-02'),

('Parth Desai', 'parth.desai@gmail.com', '2026-02-08'),

('Sneha Nair', 'sneha.nair@gmail.com', '2026-02-14'),

('Harsh Modi', 'harsh.modi@gmail.com', '2026-02-20'),

('Nidhi Kapoor', 'nidhi.kapoor@gmail.com', '2026-02-28'), ('Karan

Trivedi', 'karan.trivedi@gmail.com', '2026-03-05'),

('Isha Gupta', 'isha.gupta@gmail.com', '2026-03-12'),

('Rahul Soni', 'rahual.soni@gmail.com', '2026-03-20'),

('Neha Singh', 'neha.singh@gmail.com', '2026-03-28'),

('Vikas Chauhan', 'vikas.chauhan@gmail.com', '2026-04-05');

INSERT INTO restaurants (name, cuisine) VALUES

('Royal Spice', 'Indian'),

('Italian Oven', 'Italian'),

('Beijing Bowl', 'Chinese'),

('Burger Factory', 'American'),

('Mexican Fiesta', 'Mexican'),

('Tokyo Express', 'Japanese'),

('Punjabi Dhaba', 'North Indian'),
('Cafe Delight', 'Continental');

INSERT INTO orders (customer_id, restaurant_id, total, status, created_at) VALUES
(1,1,540.00,'completed','2026-01-06 12:30:00'),
(1,3,320.00,'completed','2026-01-18 19:20:00'),

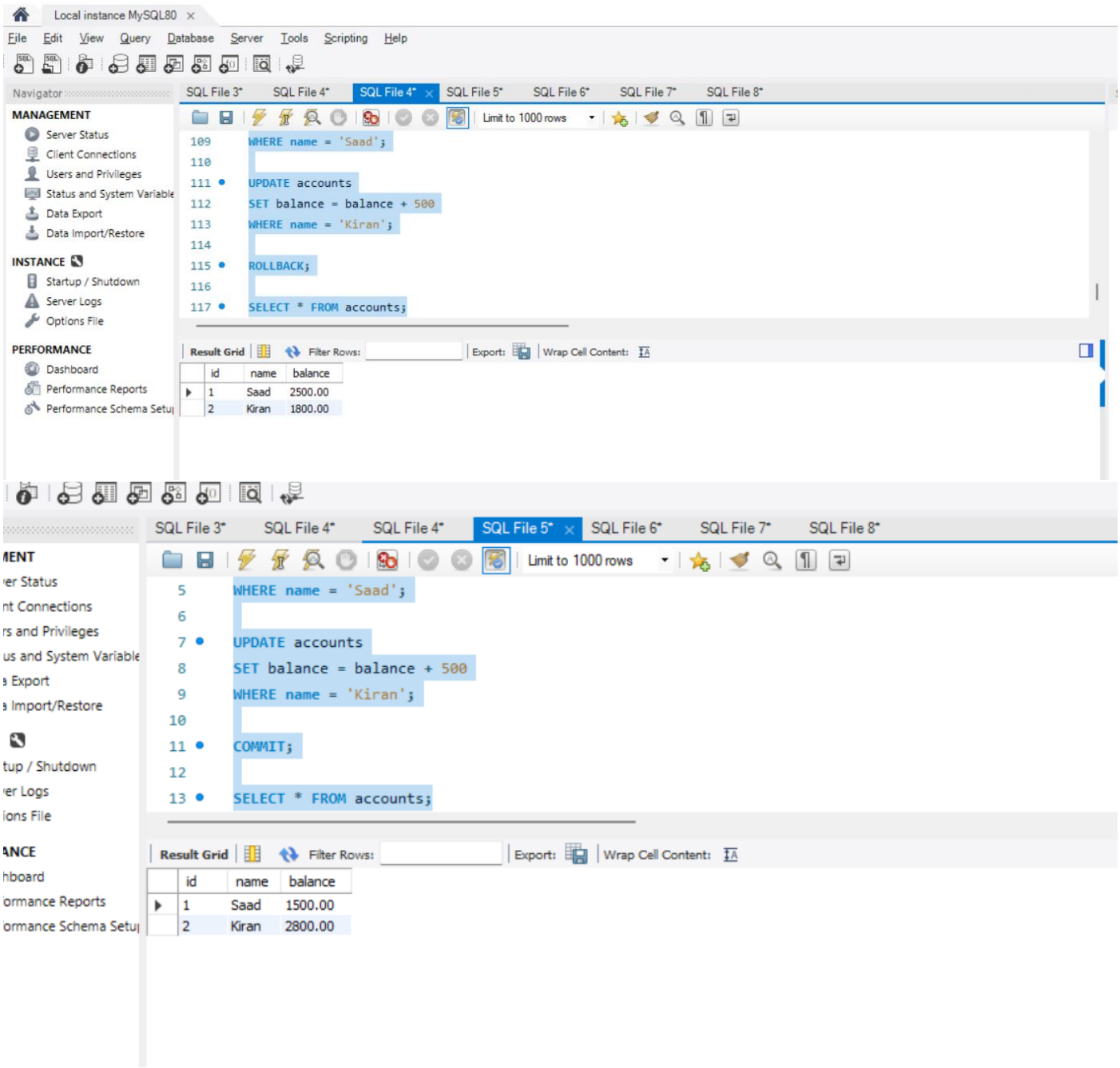
(1,6,890.00,'pending','2026-02-14 18:00:00'),
(2,2,760.00,'completed','2026-01-10 13:15:00'),
(2,5,410.00,'completed','2026-02-01 20:10:00'),
(2,8,290.00,'cancelled','2026-03-05 12:40:00'),
(3,4,650.00,'completed','2026-01-15 19:30:00'),
(3,1,430.00,'completed','2026-02-12 12:10:00'),
(3,7,980.00,'completed','2026-03-18 20:00:00'),
(4,6,1100.00,'completed','2026-01-20 18:45:00'),
(4,3,470.00,'pending','2026-02-20 13:20:00'),
(4,2,620.00,'completed','2026-03-25 19:15:00'),
(5,5,360.00,'completed','2026-01-28 12:30:00'),
(5,8,820.00,'completed','2026-02-26 18:50:00'),
(5,4,510.00,'completed','2026-03-30 20:30:00'),
(6,7,720.00,'completed','2026-02-05 13:10:00'),
(6,1,580.00,'completed','2026-03-02 19:00:00'),
(6,6,930.00,'pending','2026-04-04 18:40:00'),
(7,2,680.00,'completed','2026-02-10 12:45:00'),
(7,5,440.00,'completed','2026-03-08 20:00:00'),
(7,3,390.00,'cancelled','2026-04-08 13:00:00'),
(8,4,560.00,'completed','2026-02-18 19:40:00'),
(8,7,1250.00,'completed','2026-03-12 21:00:00'),
(8,1,470.00,'completed','2026-04-10 12:20:00'),
(9,6,880.00,'completed','2026-02-24 18:15:00'),
(9,2,630.00,'pending','2026-03-16 20:45:00'),
(9,8,350.00,'completed','2026-04-12 13:30:00'),
(10,3,420.00,'completed','2026-03-01 12:50:00'),
(10,5,710.00,'completed','2026-03-20 18:30:00'),
(10,7,980.00,'completed','2026-04-14 19:20:00'),
(11,1,520.00,'completed','2026-03-06 13:40:00'),
(11,4,610.00,'pending','2026-03-28 20:15:00'), (11,6,790.00,'completed','2026-04-16 18:00:00'),
(12,8,450.00,'completed','2026-03-15 12:30:00'),
(12,2,690.00,'completed','2026-04-01 19:10:00'),
(12,5,380.00,'completed','2026-04-18 13:20:00');

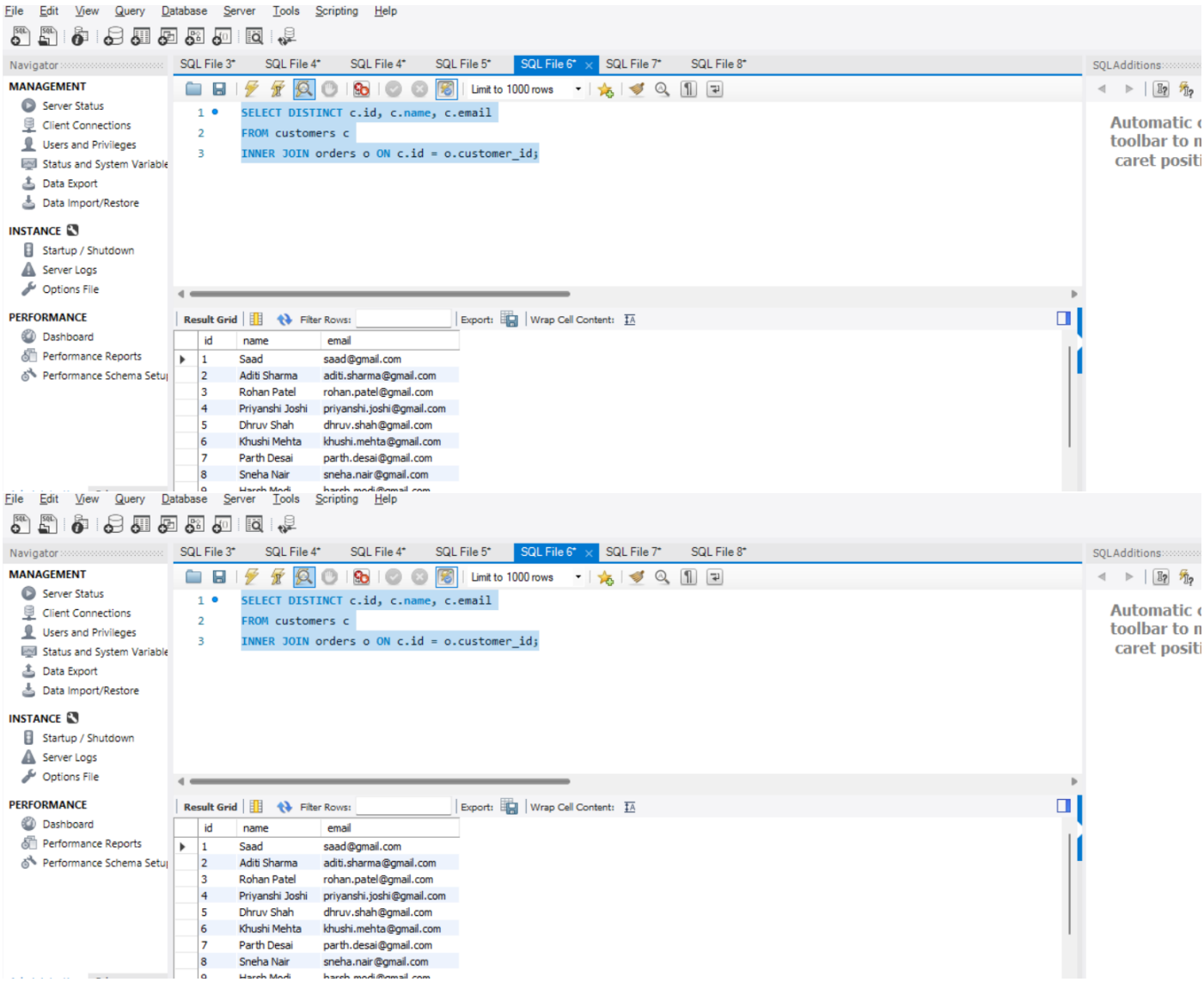
```
INSERT INTO accounts (name, balance) VALUES
('saad', 2500.00),
('Kiran', 1800.00);
```

Part – 2

TASK 1 Transactions (ACID)

a)





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SQL File 3* SQL File 4* SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8*
Limit to 1000 rows
SELECT DISTINCT c.id, c.name, c.email
FROM customers c
INNER JOIN orders o ON c.id = o.customer_id;
SQLAdditions
Automatic
toolbar to
caret posit
Result Grid Filter Rows: Export: Wrap Cell Content:
id name email
1 Saad saad@gmail.com
2 Aditi Sharma aditi.sharma@gmail.com
3 Rohan Patel rohan.patel@gmail.com
4 Priyanshi Joshi priyanshi.joshi@gmail.com
5 Dhruv Shah dhruv.shah@gmail.com
6 Khushi Mehta khushi.mehta@gmail.com
7 Parth Desai parth.desai@gmail.com
8 Sneha Nair sneha.nair@gmail.com

```sql
SELECT c.id, c.name, c.email
FROM customers c
LEFT JOIN orders o ON c.id = o.customer_id
WHERE o.id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	name	email
13	Rahul Soni	rahual.soni@gmail.com
14	Neha Singh	neha.singh@gmail.com
15	Vikas Chauhan	vikas.chauhan@gmail.com

Result 2

```sql
EXPLAIN SELECT * FROM orders
WHERE created_at = '2026-02-14 18:00:00';

CREATE INDEX idx_created_at ON orders(created_at);

EXPLAIN SELECT * FROM orders
WHERE created_at = '2026-02-14 18:00:00';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	orders	NULL	ref	idx_created_at	idx_created_at	6	const	1	100.00	NULL

Schema SQL

```
(10,3,420.00,'completed','2026-03-01 12:50:00'),
(10,5,710.00,'completed','2026-03-20 18:30:00'),
(10,7,980.00,'completed','2026-04-14 19:20:00'),
(11,1,520.00,'completed','2026-03-06 13:40:00'),
(11,4,610.00,'pending','2026-03-28 20:15:00'),
(11,6,790.00,'completed','2026-04-16 18:00:00'),
(12,8,450.00,'completed','2026-03-15 12:30:00'),
(12,2,690.00,'completed','2026-04-01 19:10:00'),
(12,5,380.00,'completed','2026-04-18 13:20:00');

INSERT INTO accounts (name, balance) VALUES
('Yuvraj', 2500.00),
('Kiran', 1800.00);
```

Text to DDL

Query SQL

```
START TRANSACTION;

UPDATE accounts
SET balance = balance - 500
WHERE name = 'Yuvraj';

UPDATE accounts
SET balance = balance + 500
WHERE name = 'Kiran';

ROLLBACK;
SELECT * FROM accounts;
```

Results

Copy as Markdown

There are no results to be displayed.

Query #5 Execution time: 0.16ms

id	name	balance
1	Yuvraj	2500.0
2	Kiran	1800.0

b)

Schema SQL

```
(10,3,420.00,'completed','2026-03-01 12:50:00'),
(10,5,710.00,'completed','2026-03-20 18:30:00'),
(10,7,980.00,'completed','2026-04-14 19:20:00'),
(11,1,520.00,'completed','2026-03-06 13:40:00'),
(11,4,610.00,'pending','2026-03-28 20:15:00'),
(11,6,790.00,'completed','2026-04-16 18:00:00'),
(12,8,450.00,'completed','2026-03-15 12:30:00'),
(12,2,690.00,'completed','2026-04-01 19:10:00'),
(12,5,380.00,'completed','2026-04-18 13:20:00');

INSERT INTO accounts (name, balance) VALUES
('Yuvraj', 2500.00),
('Kiran', 1800.00);
```

Text to DDL

Query SQL

```
START TRANSACTION;

UPDATE accounts
SET balance = balance - 500
WHERE name = 'Yuvraj';

UPDATE accounts
SET balance = balance + 500
WHERE name = 'Kiran';

COMMIT;

SELECT * FROM accounts;
```

Results

Copy as Markdown

There are no results to be displayed.

Query #9 Execution time: 0.14ms

id	name	balance
1	Yuvraj	2000.0
2	Kiran	2300.0

c)

Atomicity is the ACID property that ensures the money is never lost during a transfer. It makes sure that either both operations (debit and credit) happen successfully, or neither happens. If any error occurs, the transaction is rolled back, so both account balances remain unchanged.

TASK 2 Joins

a)

Schema SQL

```sql
CREATE TABLE customers (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100),
    email VARCHAR(100) UNIQUE,
    signup_date DATE
);

CREATE TABLE restaurants (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100),
    cuisine VARCHAR(50)
);
```

Text to DDL

Query SQL

```sql
WHERE name = 'Kiran';

COMMIT;

SELECT * FROM accounts;

SELECT DISTINCT
    c.id,
    c.name,
    c.email
FROM customers c
INNER JOIN orders o
ON c.id = o.customer_id;
```

Results

Copy as Markdown

Query #10 Execution time: 0.31ms

id	name	email
1	Yuvraj Verma	yuvraj.verma@gmail.com
2	Aditi Sharma	aditi.sharma@gmail.com
3	Rohan Patel	rohan.patel@gmail.com
4	Priyanshi Joshi	priyanshi.joshi@gmail.com

b)

Schema SQL

```sql
('Parth Desai', 'parth.desai@gmail.com', '2026-02-08'),
('Sneha Nair', 'sneha.nair@gmail.com', '2026-02-14'),
('Harsh Modi', 'harsh.modi@gmail.com', '2026-02-20'),
('Nidhi Kapoor', 'nidhi.kapoor@gmail.com', '2026-02-28'),
('Karan Trivedi', 'karan.trivedi@gmail.com', '2026-03-05'),
('Isha Gupta', 'isha.gupta@gmail.com', '2026-03-12'),
('Rahul Soni', 'rahual.soni@gmail.com', '2026-03-20'),
('Neha Singh', 'neha.singh@gmail.com', '2026-03-28'),
('Vikas Chauhan', 'vikas.chauhan@gmail.com', '2026-04-05');

INSERT INTO restaurants (name, cuisine) VALUES
('Royal Spice', 'Indian'),
('Italian Oven', 'Italian'),
('Beijing Bowl', 'Chinese'),
```

Text to DDL

Query SQL

```sql
    c.name,
    c.email
FROM customers c
INNER JOIN orders o
ON c.id = o.customer_id;

SELECT
    c.id,
    c.name,
    c.email
FROM customers c
LEFT JOIN orders o
ON c.id = o.customer_id
WHERE o.id IS NULL;
```

Results

Copy as Markdown

Query #11 Execution time: 0.19ms

id	name	email
13	Rahul Soni	rahual.soni@gmail.com
14	Neha Singh	neha.singh@gmail.com
15	Vikas Chauhan	vikas.chauhan@gmail.com

c)

A marketing team prefers query (b) because it also shows customers who have never placed an order, so they can target them with special offers or promotions.

TASK 3 Indexing

a)

We are using the **status** column from the orders table, which currently does not have an index. b)

c) & d)

e) Before adding the index, MySQL had to perform a full table scan examining all 36 rows; after indexing, it performed an instant index lookup examining only the 5 matching rows, significantly cutting down execution time.

TASK 4 Query Optimization

a)

Schema SQL

```
59
60 INSERT INTO orders (customer_id, restaurant_id, total, status, created_at) VALUES
61 (1,1,540.00,'completed','2026-01-06 12:30:00'),
62 (1,3,320.00,'completed','2026-01-18 19:20:00'),
63 (1,6,890.00,'pending','2026-02-14 18:00:00'),
64 (2,2,760.00,'completed','2026-01-10 13:15:00'),
65 (2,5,410.00,'completed','2026-02-01 20:10:00'),
66 (2,8,290.00,'cancelled','2026-03-05 12:40:00'),
67 (3,4,650.00,'completed','2026-01-15 19:30:00'),
68 (3,1,430.00,'completed','2026-02-12 12:10:00'),
69 (3,7,980.00,'completed','2026-03-18 20:00:00'),
70 (4,6,1100.00,'completed','2026-01-20 18:45:00'),
71 (4,3,470.00,'pending','2026-02-20 13:20:00'),
72 (4,2,620.00,'completed','2026-03-25 19:15:00'),
```

Text to DDL

Query SQL

```
1 SELECT o.*
2 FROM orders o
3 INNER JOIN customers c
4 ON o.customer_id = c.id
5 WHERE c.signup_date > '2026-02-01';
```

Results

Copy as Markdown

Query #1 Execution time: 0.42ms

id	customer_id	restaurant_id	total	status	created_at
16	6	7	720.0	completed	2026-02-05 13:10:00
17	6	1	580.0	completed	2026-03-02 19:00:00
18	6	6	930.0	pending	2026-04-04 18:40:00
19	7	2	680.0	completed	2026-02-10 12:45:00

b)

Run Save Load Example Collaborate Sign in Have any feedback?

Schema SQL

```
56 ('Punjabi Dhaba', 'North Indian'),
57 ('Cafe Delight', 'Continental');
58
59
60 INSERT INTO orders (customer_id, restaurant_id, total, status, created_at) VALUES
61 (1,1,540.00,'completed','2026-01-06 12:30:00'),
62 (1,3,320.00,'completed','2026-01-18 19:20:00'),
63 (1,6,890.00,'pending','2026-02-14 18:00:00'),
64 (2,2,760.00,'completed','2026-01-10 13:15:00'),
65 (2,5,410.00,'completed','2026-02-01 20:10:00'),
66 (2,8,290.00,'cancelled','2026-03-05 12:40:00'),
67 (3,4,650.00,'completed','2026-01-15 19:30:00'),
68 (3,1,430.00,'completed','2026-02-12 12:10:00'),
69 (3,7,980.00,'completed','2026-03-18 20:00:00'),
```

Text to DDL

Query SQL

```
1 SELECT
2   o.id,
3   o.restaurant_id,
4   o.status,
5   o.total
6 FROM orders o
7 INNER JOIN customers c
8 ON o.customer_id = c.id
9 WHERE c.signup_date > '2026-02-01';
```

Results

Copy as Markdown

Query #1 Execution time: 0.37ms

id	restaurant_id	status	total
16	7	completed	720.0
17	1	completed	580.0
18	6	pending	930.0
19	2	completed	680.0

c)

Schema SQL

```
('Punjabi Dhaba', 'North Indian'),
('Cafe Delight', 'Continental');


INSERT INTO orders (customer_id, restaurant_id, total, status, created_at) VALUES
(1,1,540.00,'completed','2026-01-06 12:30:00'),
(1,3,320.00,'completed','2026-01-18 19:20:00'),
(1,6,890.00,'pending','2026-02-14 18:00:00'),
(2,2,760.00,'completed','2026-01-10 13:15:00'),
(2,5,410.00,'completed','2026-02-01 20:10:00'),
(2,8,290.00,'cancelled','2026-03-05 12:40:00'),
(3,4,650.00,'completed','2026-01-15 19:30:00'),
(3,1,430.00,'completed','2026-02-12 12:10:00'),
(3,7,980.00,'completed','2026-03-18 20:00:00'),
```

Text to DDL

Query SQL

```
EXPLAIN ANALYZE
SELECT *FROM orders WHERE customer_id IN
(
    SELECT id FROM customers
    WHERE signup_date > '2026-02-01'
);

EXPLAIN ANALYZE
SELECT o.id, o.restaurant_id, o.status,o.total
FROM orders o
INNER JOIN customers c
ON o.customer_id = c.id
WHERE c.signup_date > '2026-02-01';
```

Results

Copy as Markdown

-> Nested loop inner join (cost=3.28 rows=4.67) (actual time=0.0403..0.0833 rows=21 loops=1) -> Filter: (customers.signup_date > DATE'2026-02-01') (cost=1.65 rows=4.67) (actual time=0.0185..0.0239 rows=10 loops=1) -> Table scan on customers (cost=1.65 rows=14) (actual time=0.0152..0.0206 rows=15 loops=1) -> Index lookup on orders using customer_id (customer_id = customers.id) (cost=0.271 rows=1) (actual time=0.00457..0.00514 rows=2.1 loops=10)

Query #2 Execution time: 0.3ms

EXPLAIN
-> Nested loop inner join (cost=3.28 rows=4.67) (actual time=0.02..0.0607 rows=21 loops=1) -> Filter: (c.signup_date > DATE'2026-02-01') (cost=1.65 rows=4.67) (actual time=0.0102..0.0151 rows=10 loops=1) -> Table scan on c (cost=1.65 rows=14) (actual time=0.00811..0.0129 rows=15 loops=1) -> Index lookup on o using customer_id (customer_id = c.id) (cost=0.271 rows=1) (actual time=0.00367..0.0042 rows=2.1 loops=10)

TASK 5 The Customer Profile Page

a)

Schema SQL

```
('Punjabi Dhaba', 'North Indian'),
('Cafe Delight', 'Continental');


INSERT INTO orders (customer_id, restaurant_id, total, status, created_at) VALUES
(1,1,540.00,'completed','2026-01-06 12:30:00'),
(1,3,320.00,'completed','2026-01-18 19:20:00'),
(1,6,890.00,'pending','2026-02-14 18:00:00'),
(2,2,760.00,'completed','2026-01-10 13:15:00'),
(2,5,410.00,'completed','2026-02-01 20:10:00'),
(2,8,290.00,'cancelled','2026-03-05 12:40:00'),
(3,4,650.00,'completed','2026-01-15 19:30:00'),
(3,1,430.00,'completed','2026-02-12 12:10:00'),
(3,7,980.00,'completed','2026-03-18 20:00:00'),
```

Text to DDL

Query SQL

```
SELECT
    r.name AS restaurant_name,
    o.total,
    o.status,
    o.created_at
FROM orders o
INNER JOIN restaurants r
ON o.restaurant_id = r.id
WHERE o.customer_id = 1
ORDER BY o.created_at DESC
LIMIT 5;
```

Results

Copy as Markdown

Query #1 Execution time: 0.33ms

restaurant_name	total	status	created_at
Tokyo Express	890.0	pending	2026-02-14 18:00:00
Beijing Bowl	320.0	completed	2026-01-18 19:20:00
Royal Spice	540.0	completed	2026-01-06 12:30:00

b)

Schema SQL

```
('Punjabi Dhaba', 'North Indian'),
('Cafe Delight', 'Continental');


INSERT INTO orders (customer_id, restaurant_id, total, status, created_at) VALUES
(1,1,540.00,'completed','2026-01-06 12:30:00'),
(1,3,320.00,'completed','2026-01-18 19:20:00'),
(1,6,890.00,'pending','2026-02-14 18:00:00'),
(2,2,760.00,'completed','2026-01-10 13:15:00'),
(2,5,410.00,'completed','2026-02-01 20:10:00'),
(2,8,290.00,'cancelled','2026-03-05 12:40:00'),
(3,4,650.00,'completed','2026-01-15 19:30:00'),
(3,1,430.00,'completed','2026-02-12 12:10:00'),
(3,7,980.00,'completed','2026-03-18 20:00:00'),
```

Text to DDL

Query SQL

```
EXPLAIN ANALYZE
SELECT
    r.name AS restaurant_name,
    o.total,
    o.status,
    o.created_at
FROM orders o
JOIN restaurants r
ON o.restaurant_id = r.id
WHERE o.customer_id = 1
ORDER BY o.created_at DESC
LIMIT 5;
```

Results

Copy as Markdown

Query #1 Execution time: 17.05ms

EXPLAIN

-> Limit: 5 row(s) (cost=2.1 rows=3) (actual time=0.0502..0.0546 rows=3 loops=1) -> Nested loop inner join (cost=2.1 rows=3) (actual time=0.0466..0.0507 rows=3 loops=1) -> Sort: o.created_at DESC (cost=1.05 rows=3) (actual time=0.0341..0.0346 rows=3 loops=1) -> Filter: (o.restaurant_id is not null) (cost=1.05 rows=3) (actual time=0.0177..0.0198 rows=3 loops=1) -> Index lookup on o using customer_id (customer_id = 1) (cost=1.05 rows=3) (actual time=0.0165..0.0183 rows=3 loops=1) -> Single-row index lookup on r using PRIMARY (id = o.restaurant_id) (cost=0.283 rows=1) (actual time=0.00468..0.00473 rows=1 loops=3)

c) & d)

Schema SQL

```
('Punjabi Dhaba', 'North Indian'),
('Cafe Delight', 'Continental');


INSERT INTO orders (customer_id, restaurant_id, total, status, created_at) VALUES
(1,1,540.00,'completed','2026-01-06 12:30:00'),
(1,3,320.00,'completed','2026-01-18 19:20:00'),
(1,6,890.00,'pending','2026-02-14 18:00:00'),
(2,2,760.00,'completed','2026-01-10 13:15:00'),
(2,5,410.00,'completed','2026-02-01 20:10:00'),
(2,8,290.00,'cancelled','2026-03-05 12:40:00'),
(3,4,650.00,'completed','2026-01-15 19:30:00'),
(3,1,430.00,'completed','2026-02-12 12:10:00'),
(3,7,980.00,'completed','2026-03-18 20:00:00'),
```

Text to DDL

Query SQL

```
CREATE INDEX idx_customer_created
ON orders(customer_id, created_at);

EXPLAIN ANALYZE
SELECT r.name AS restaurant_name,o.total,o.status,o.created_at
FROM orders o
JOIN restaurants r
ON o.restaurant_id = r.id
WHERE o.customer_id = 1
ORDER BY o.created_at DESC
LIMIT 5;
```

Results

Copy as Markdown

Query #1 Execution time: 18.68ms

There are no results to be displayed.

Query #2 Execution time: 0.65ms

EXPLAIN

-> Limit: 5 row(s) (cost=2.1 rows=3) (actual time=0.0434..0.0497 rows=3 loops=1) -> Nested loop inner join (cost=2.1 rows=3) (actual time=0.0394..0.0454 rows=3 loops=1) -> Filter: (o.restaurant_id is not null) (cost=1.05 rows=3) (actual time=0.0304..0.0329 rows=3 loops=1) -> Index lookup on o using idx_customer_created (customer_id = 1) (reverse) (cost=1.05 rows=3) (actual time=0.0291..0.0312 rows=3 loops=1) -> Single-row index lookup on r using PRIMARY (id = o.restaurant_id) (cost=0.283 rows=1) (actual time=0.00344..0.00348 rows=1 loops=3)

e)

Schema SQL

```
('Punjabi Dhaba', 'North Indian'),
('Cafe Delight', 'Continental');


INSERT INTO orders (customer_id, restaurant_id, total, status, created_at) VALUES
(1,1,540.00,'completed','2026-01-06 12:30:00'),
(1,3,320.00,'completed','2026-01-18 19:20:00'),
(1,6,890.00,'pending','2026-02-14 18:00:00'),
(2,2,760.00,'completed','2026-01-10 13:15:00'),
(2,5,410.00,'completed','2026-02-01 20:10:00'),
(2,8,290.00,'cancelled','2026-03-05 12:40:00'),
(3,4,650.00,'completed','2026-01-15 19:30:00'),
(3,1,430.00,'completed','2026-02-12 12:10:00'),
(3,7,980.00,'completed','2026-03-18 20:00:00'),
```

Text to DDL

Query SQL

```
SELECT
    r.name AS restaurant_name,
    o.total,
    o.status,
    o.created_at
FROM orders o
INNER JOIN restaurants r
ON o.restaurant_id = r.id
WHERE o.customer_id = 1
ORDER BY o.created_at DESC
LIMIT 5;
```

Results

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Query #1 Execution time: 0.35ms

restaurant_name	total	status	created_at
Tokyo Express	890.0	pending	2026-02-14 18:00:00
Beijing Bowl	320.0	completed	2026-01-18 19:20:00
Royal Spice	540.0	completed	2026-01-06 12:30:00

f)

We have created an index on (customer_id, created_at) because the query filters data using customer_id and sorts it using created_at. The index helps MySQL find the required records faster without scanning unnecessary data. As a result, the execution time decreased from 17.05 ms to 0.65 ms, making the query more efficient.